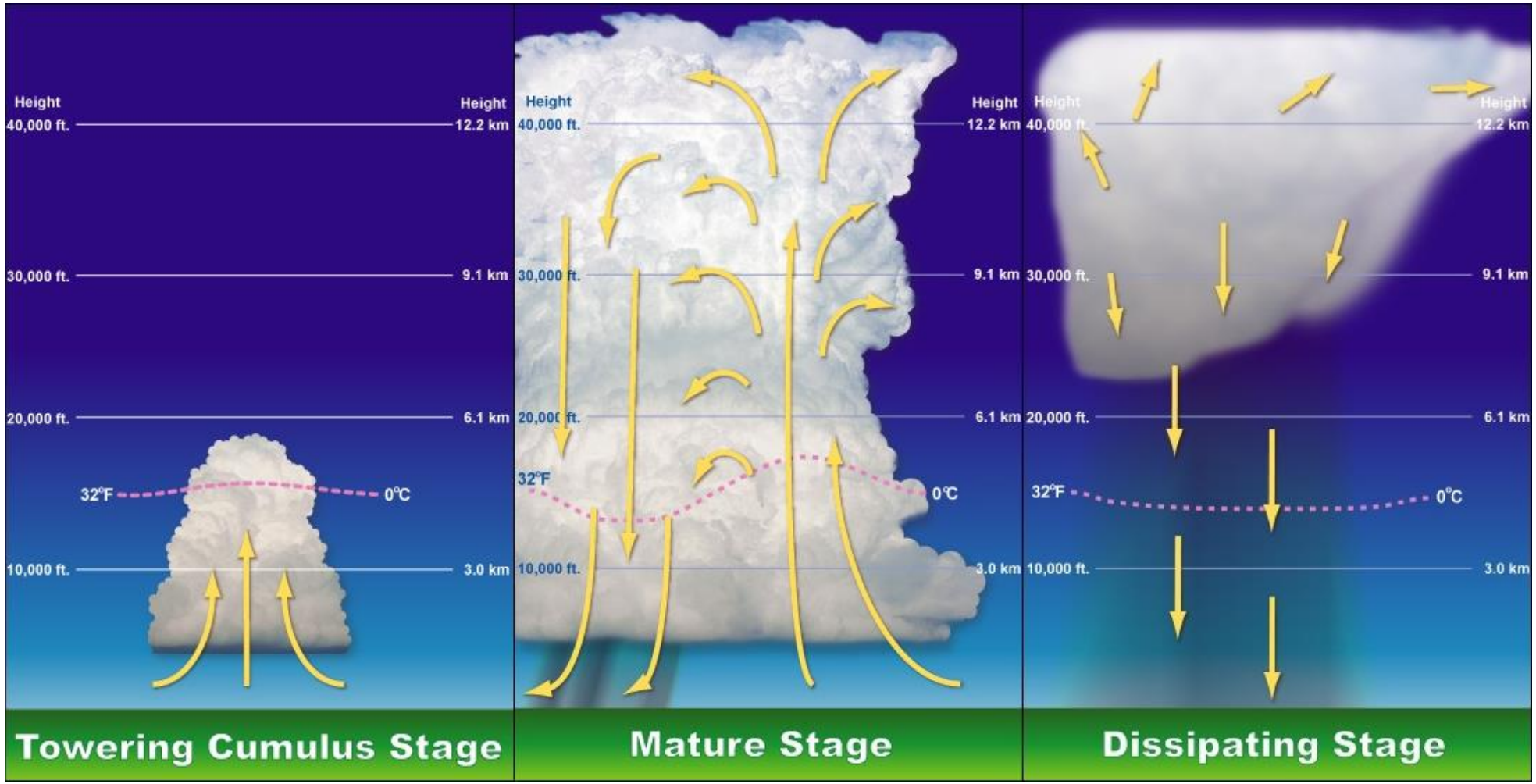
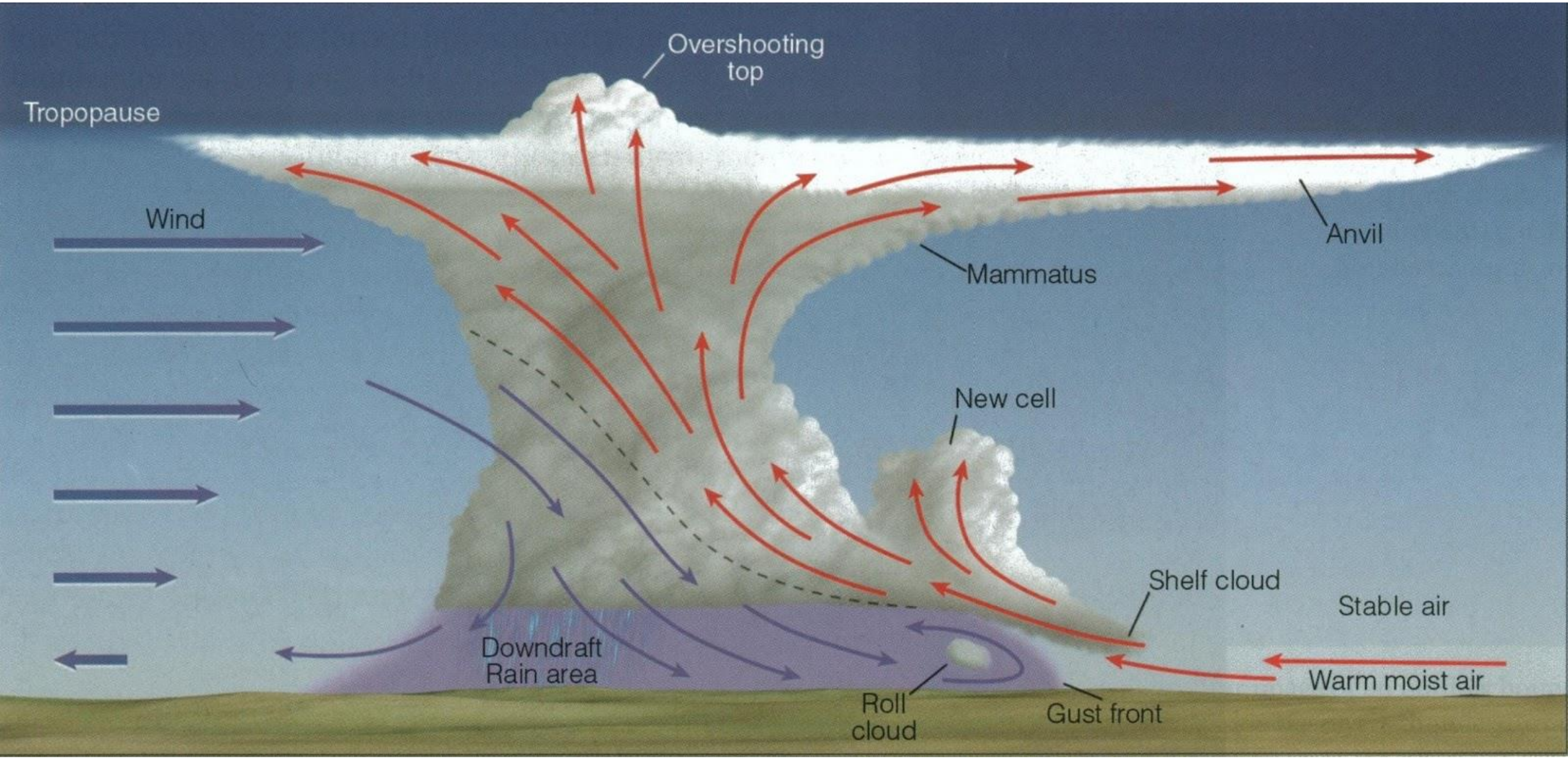
Deep convective cloud

Cloud burst

(Rain bomb ??, water bomb, weather bomb)



https://weathertogether.net/weather-101/convective-clouds/


Overshooting top
Tropopause
Wind
Anvil
Mammatus
New cell
Shelf cloud
Stable air
Downdraft
Rain area
Warm moist air
Roll cloud
Gust front

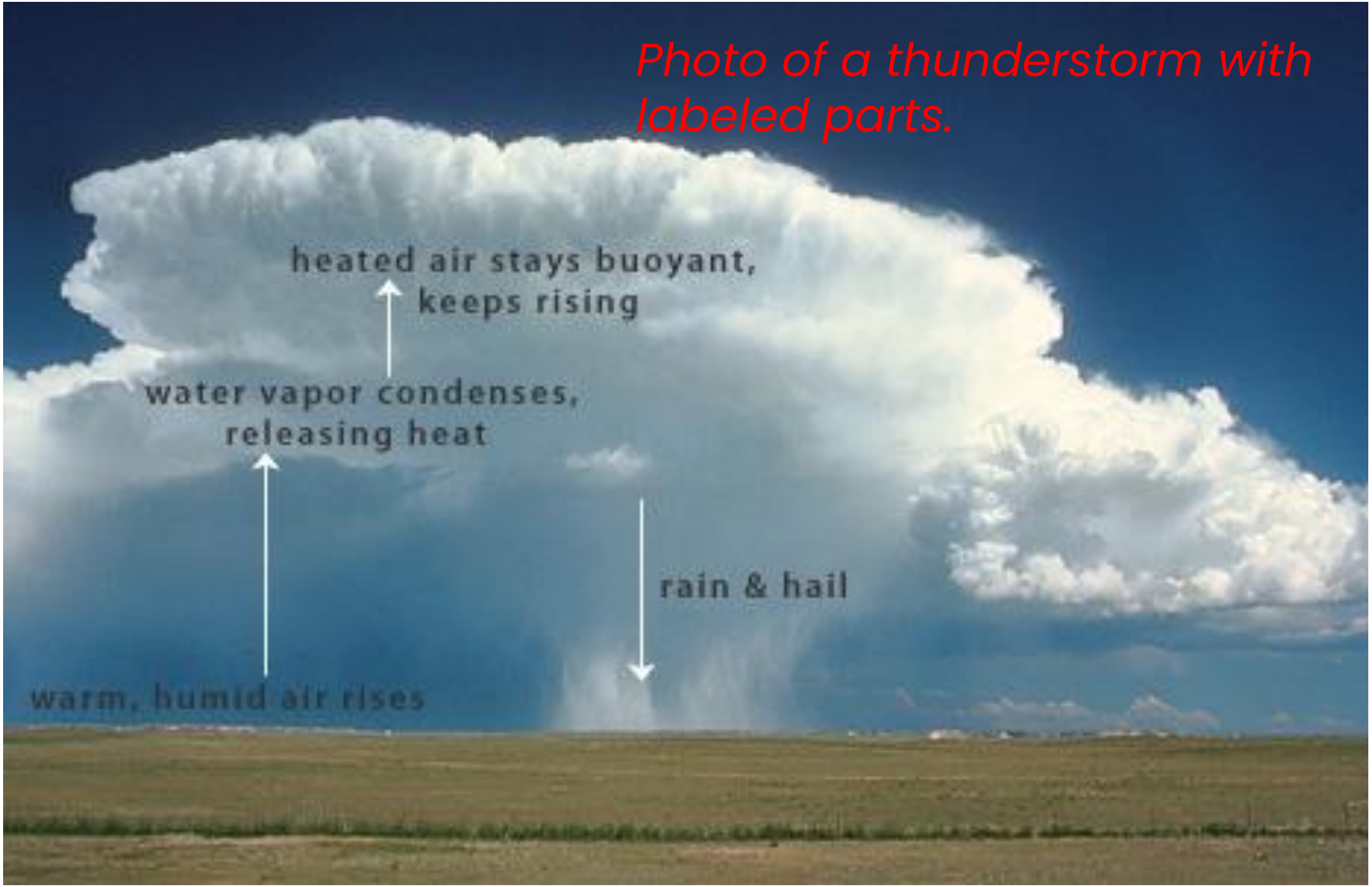
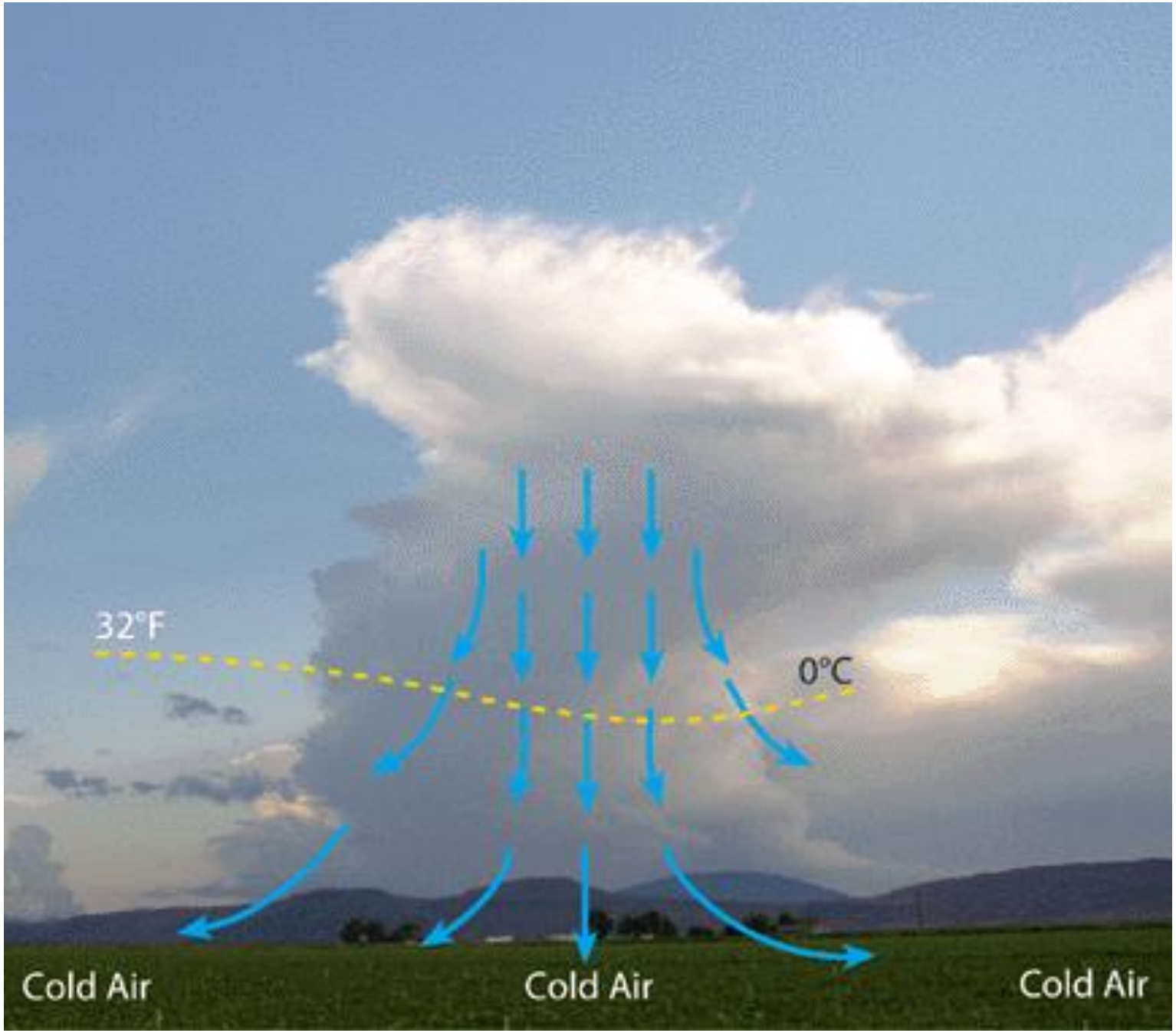


Photo of a thunderstorm with labeled parts.
heated air stays buoyant, keeps rising
water vapor condenses, releasing heat
rain & hail
warm, humid air rises



- *พายุฝนฟ้าคะนอง ในระยะที่มีการสลายตัว*
- *Dissipating thunderstorm. Retrieved from Sam Houston State University. Credit: NOAA*

https://www.iasgyan.in/daily-current-affairs/cloudbursts

คำจำกัดความ cloud burst จากสมาคมอุตุนิยมวิทยาอเมริกัน AMS

glossary.ametsoc.org/wiki/cloudburst/

AMS American Meteorological Society | Glossary of Meteorology

All Terms | Search Glossary of Meteorology | Search

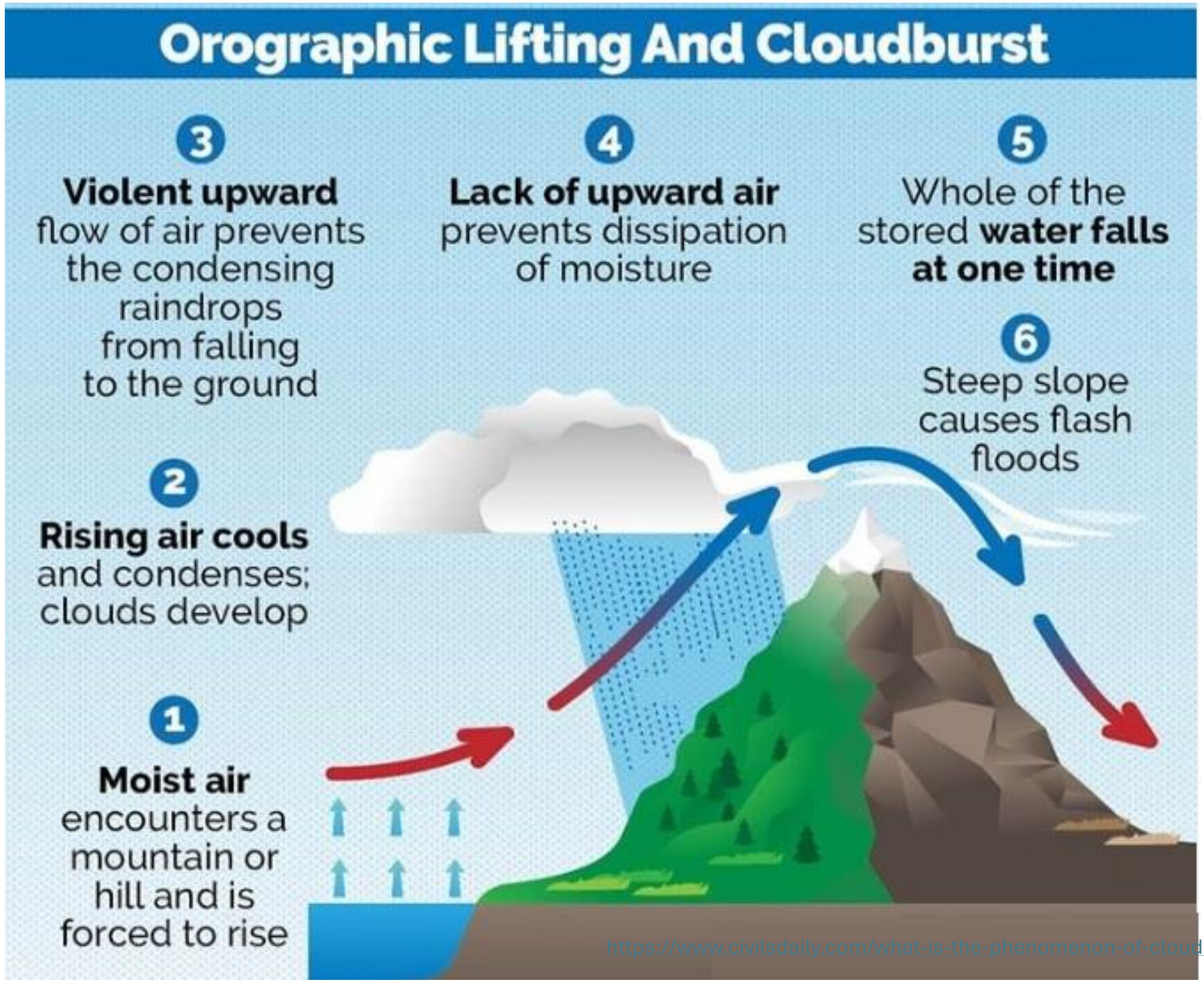
cloudburst

(*Also called* rain gush, rain gust.) In popular terminology, any sudden and heavy fall of rain, almost always of the shower type.

An unofficial criterion sometimes used specifies a rate of fall equal to or greater than 100 mm (3.94 inches) per hour.
See excessive precipitation.

Copyright 2026 American Meteorological Society (AMS). For permission to reuse any portion of this work, please contact permissions@ametsoc.org. Any use of material in this work that is determined to be "fair use" under Section 107 of the U.S. Copyright Act (17 U.S. Code § 107) or that satisfies the conditions specified in Section 108 of the U.S.Copyright Act (17 USC § 108) does not require AMS's permission. Republication, systematic reproduction, posting in electronic form, such as on a website or in a searchable database, or other uses of this material, except as exempted by the above statement, require written permission or a license from AMS. Additional details are provided in the AMS Copyright Policy statement.

https://glossary.ametsoc.org/wiki/cloudburst/

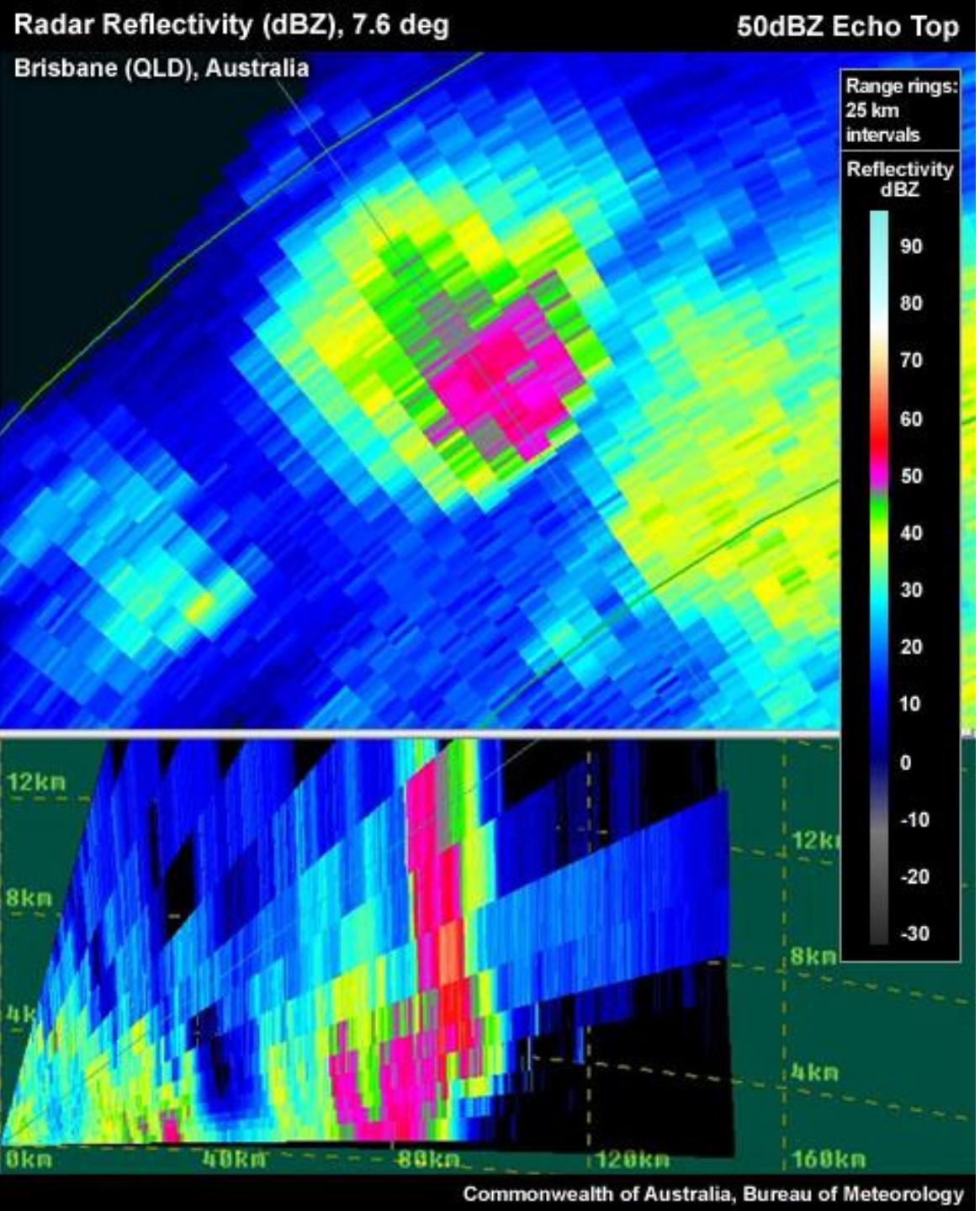


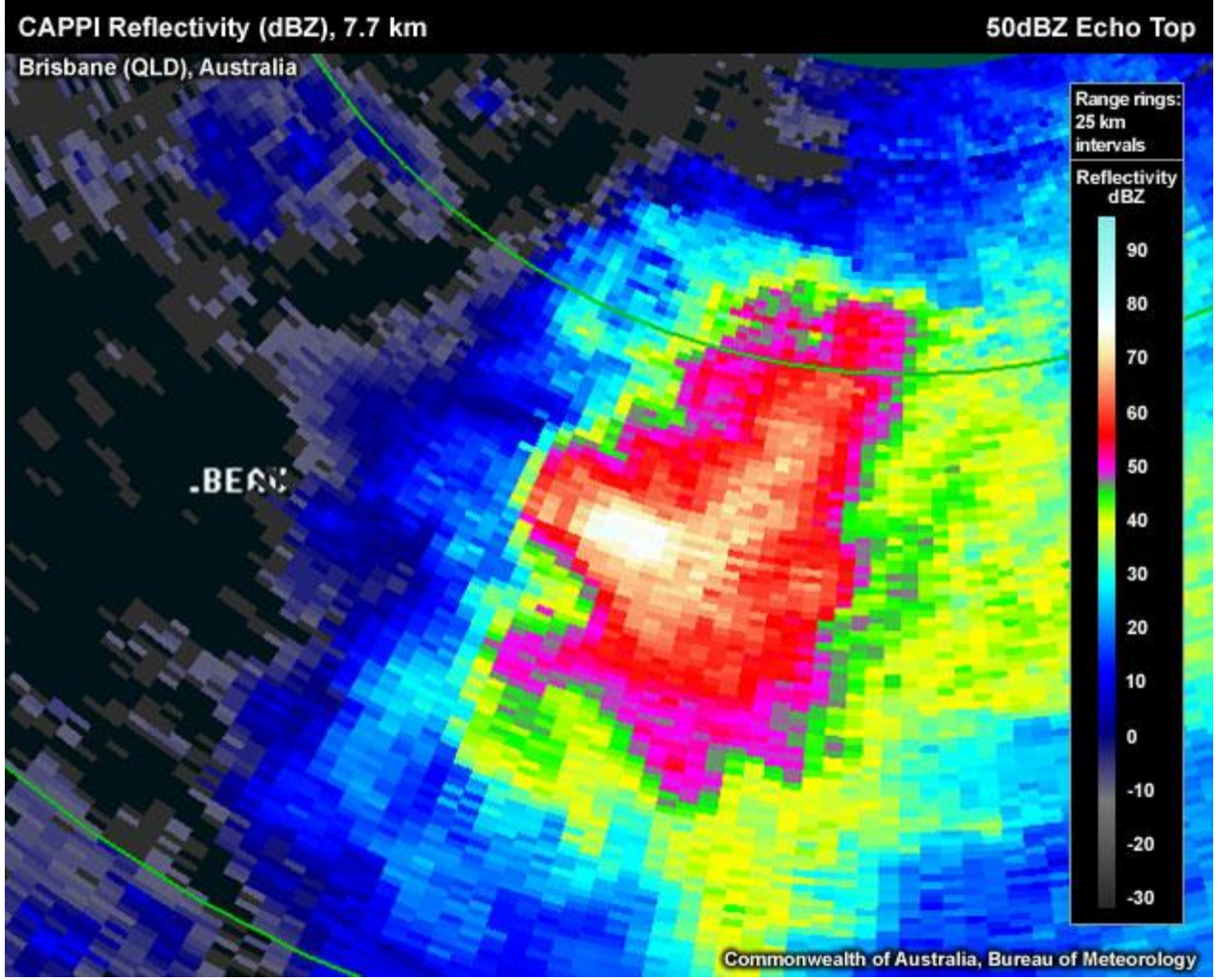
https://www.civilsdaily.com/what-is-the-phenomenon-of-cloudbursts-explain/

เหตุการณ์ฝนที่รุนแรงแต่ละชนิด

Feature	Cloudburst	Heavy Rainfall Event	Flash Flood
Definition	≥100 mm/hr. over small area (20–30 km^2)	Intense rainfall but over a larger area and/or longer duration	Sudden flooding caused by heavy rain or dam break
Scale	Highly localised	Regional or sub-regional	Local or regional
Duration	Minutes to a few hours	Hours to days	Minutes to hours
Primary cause	Orographic convection and rapid condensation	Monsoon systems, depressions	Heavy rain + topography + hydrology
Hazard link	Direct cause of flash floods and landslides	Flooding, waterlogging	May be triggered by cloudbursts or prolonged rainfall

https://www.indiawaterportal.org/climate-change/disasters/faq-understanding-cloudbursts-in-the-indian-himalayas-and-how-to-stay-safe-2





- ตรวจสอบกลุ่มเมฆในแต่ละระดับความสูง ด้วยข้อมูลเรดาร์หลายมุมยก
- ตรวจสอบระยะเวลา การเคลื่อนที่ ความเร็ว

https://www.faculty.luther.edu/~bernatzr/Courses/Sci123/comet/radar/severe_signatures/print_50dBZ.htm

เปเปอร์วิจัย cloudburst ที่ Hawaii

journals.ametsoc.org/view/journals/mwre/150/4/MWR-D-21-0028.1.xml

AMS American Meteorological Society | Journals

JOURNALS BROWSE PUBLISH SUBSCRIBE ABOUT

Previous Article Next Article

Monthly Weather Review

Volume 150: Issue 4

Sections

References

Figures

Cited By

Metrics

Editorial Type: Article

Article Type: Research Article

The Anatomy of a Series of Cloud Bursts that Eclipsed the U.S. Rainfall Record

Terrence J. Corrigan Jr. and Steven Businger

Online Publication: 07 Apr 2022

Print Publication: 01 Apr 2022

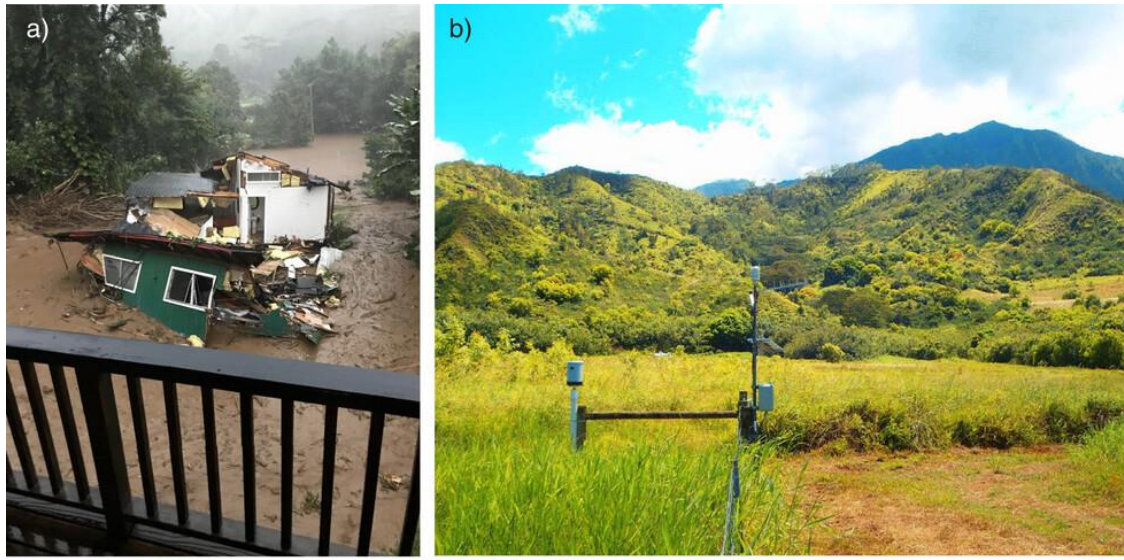
DOI: https://doi.org/10.1175/MWR-D-21-0028.1

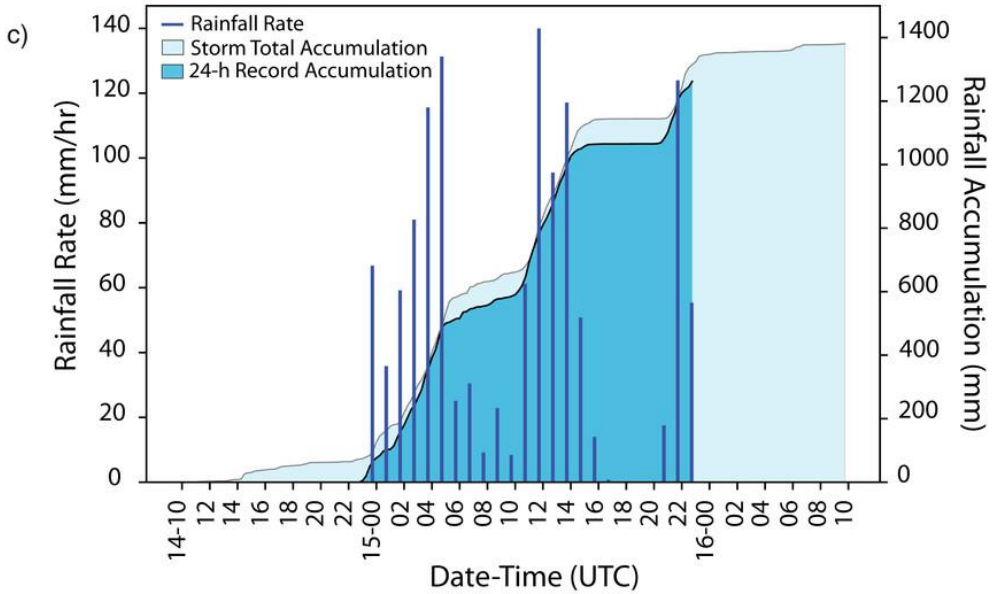
Page(s): 753–773

Cited by: 3

Article History | Download PDF | © Get Permissions

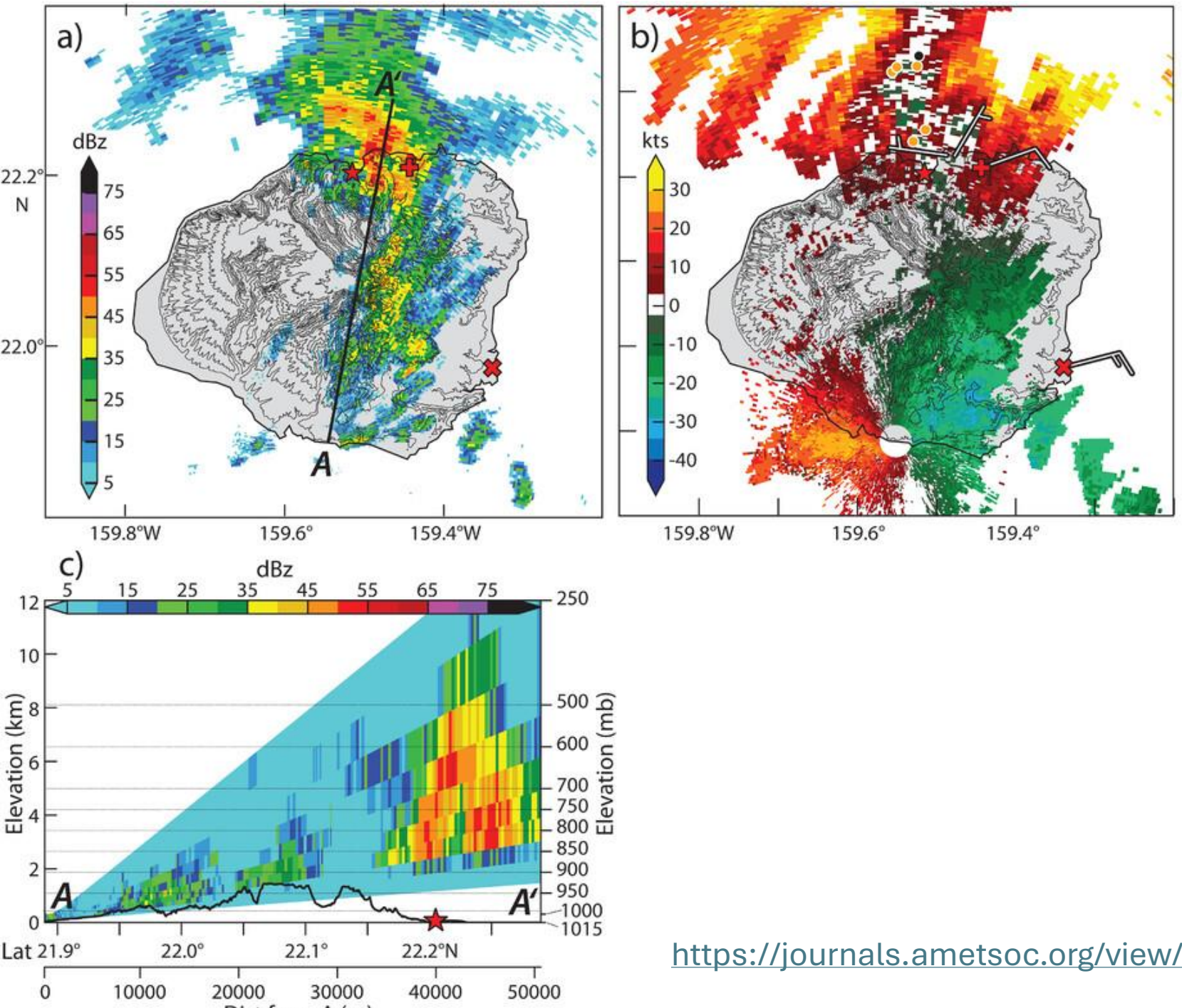
Abstract/Excerpt | Full Text | PDF





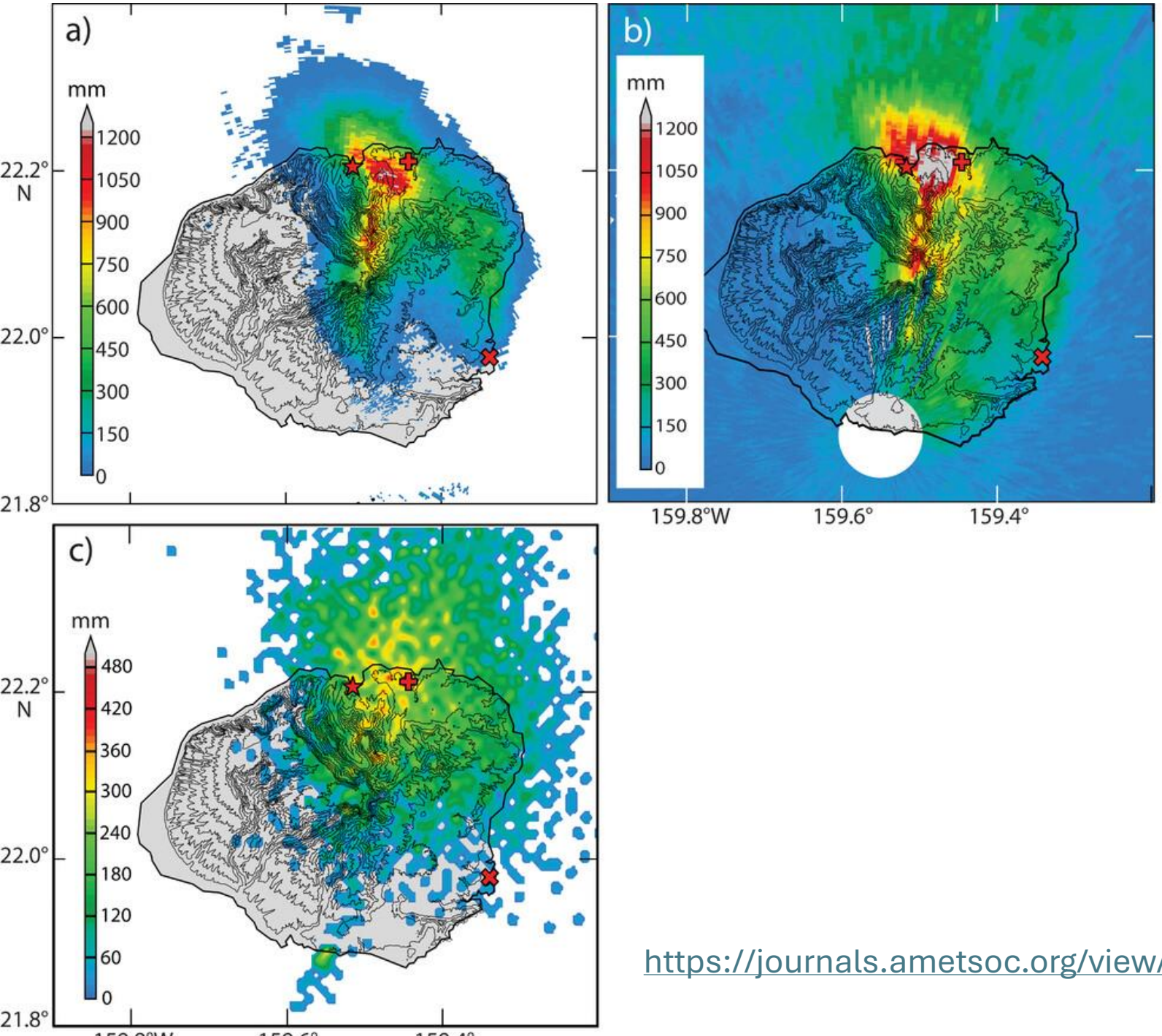
- 24-h rainfall record was broken by this storm, with 1262 mm (～50 in.)

https://journals.ametsoc.org/view/journals/mwre/150/4/MWR-D-21-0028.1.xml



- FIG. 10.
- WSR88-D data for 0423 UTC 15 Apr 2018. (a) PPI radar reflectivity at scan angle 3.8° and
- (b) PPI radial velocity at scan angle 4.8°. Vaisala GLD360 positive and negative polarity lightning strikes within a ±2-min window are plotted as black and orange circles, respectively, and available surface sustained winds are plotted as wind barbs in knots (1 kt ≈ 0.5 m s^{-1}).
- (c) RHI reflectivity and USGS high-resolution elevation (black line) from cross-section line A–A' in (a). USGS elevation contours are every 150 m for (a) and (b), and station locations in (a)–(c) are as defined in Fig. 2.

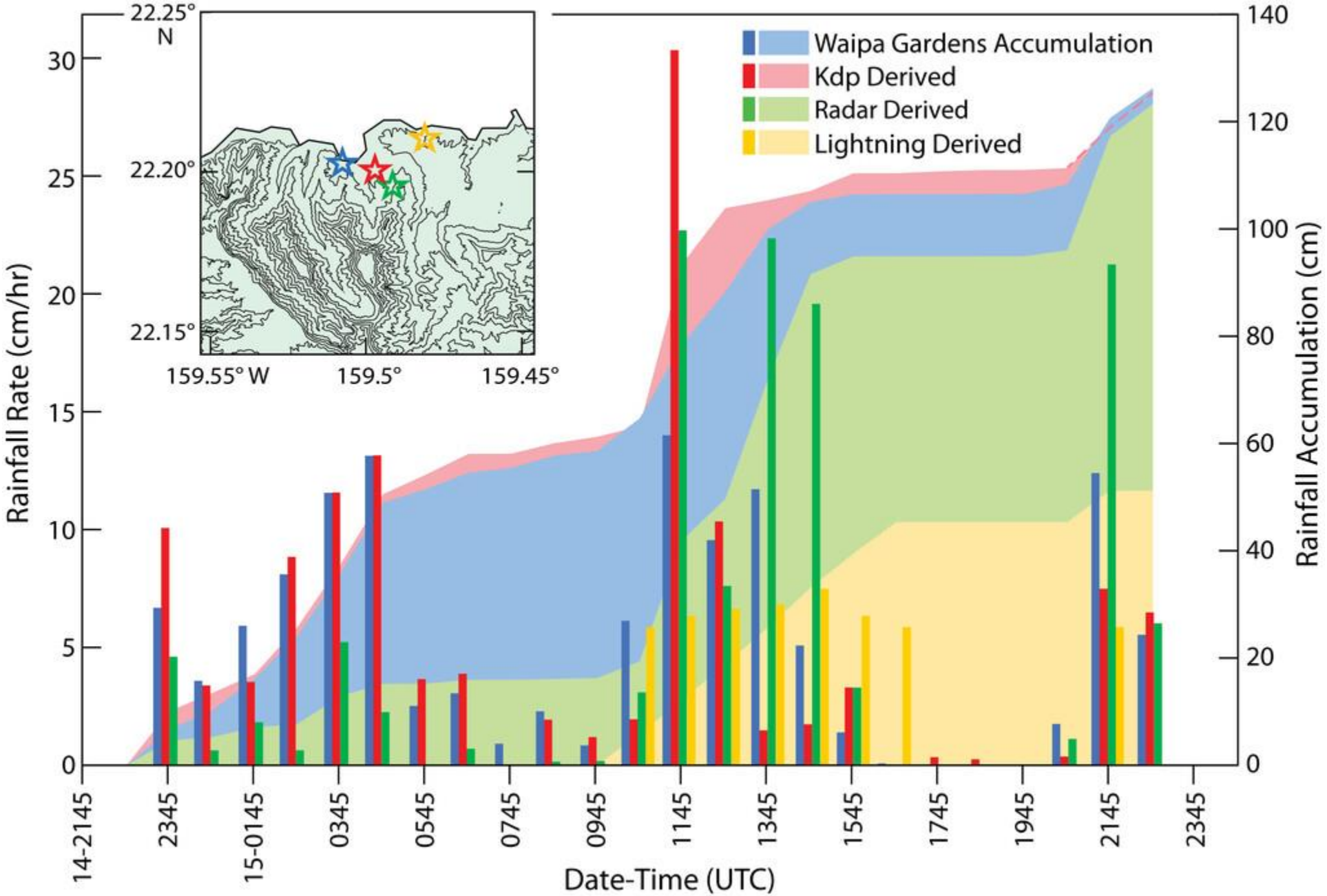
https://journals.ametsoc.org/view/journals/mwre/150/4/MWR-D-21-0028.1.xml

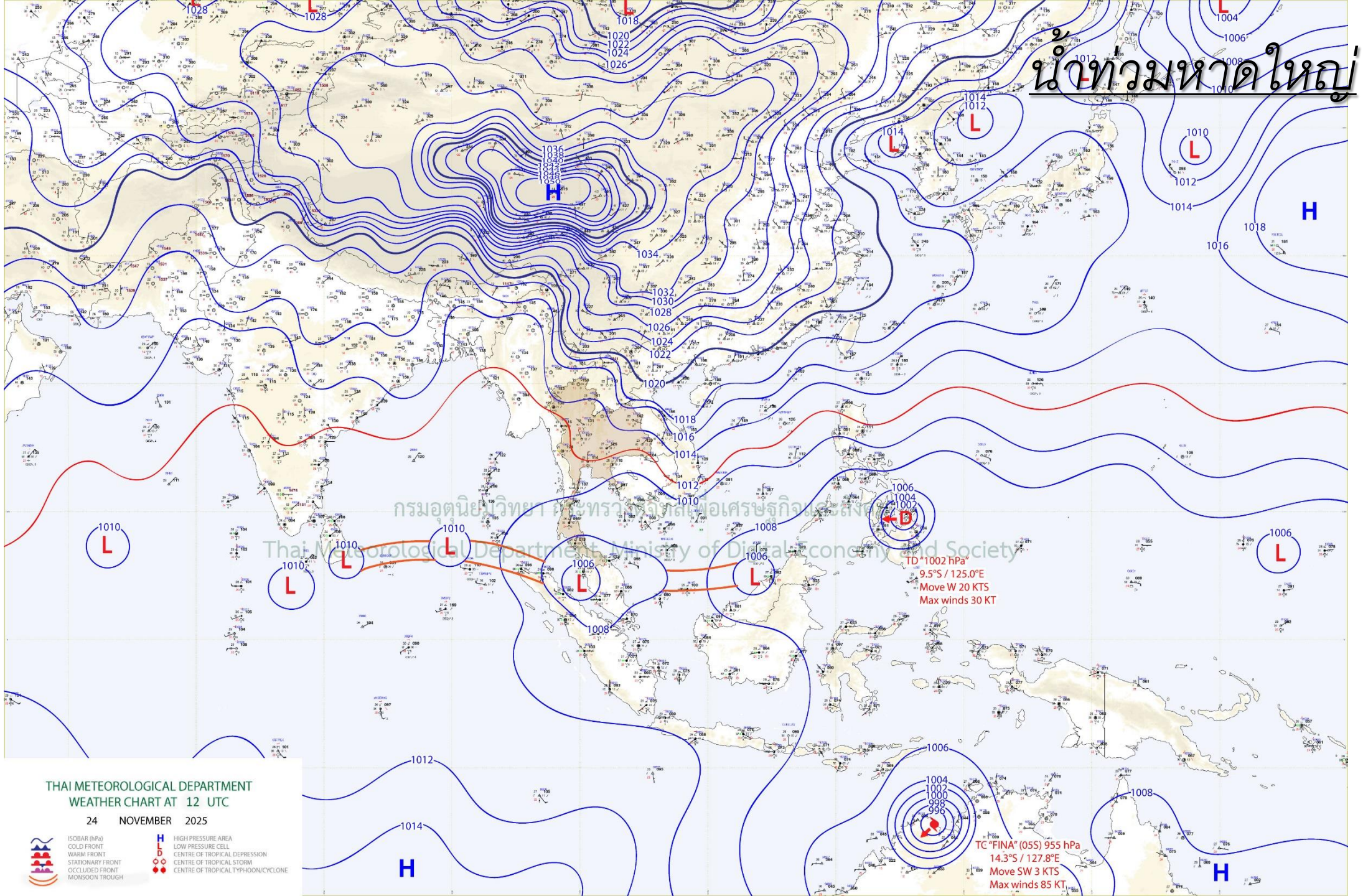


Twenty-four-hour rainfall accumulation beginning at 2245 UTC 14 Apr (a) derived from WSR-88D,

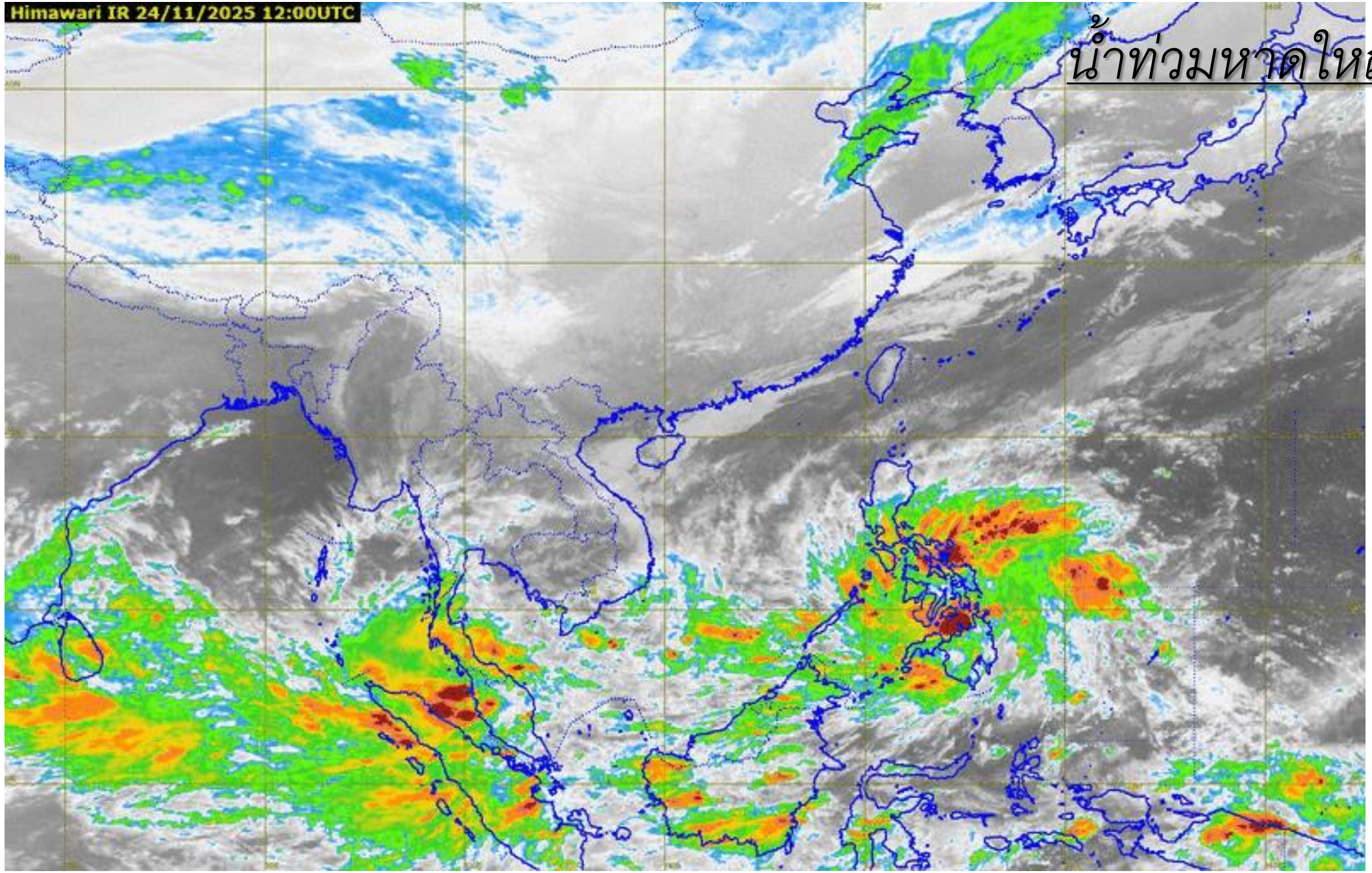
- (b) derived from WSR-88D K_{DP} accumulation, and
- (c) derived from Vaisala GLD360 accumulated lightning density (strikes per kilometer squared). USGS high-resolution elevation are contoured every 150 m. Waipā, Princeville, and Lihue stations are marked as in Fig. 2.

https://journals.ametsoc.org/view/journals/mwre/150/4/MWR-D-21-0028.1.xml

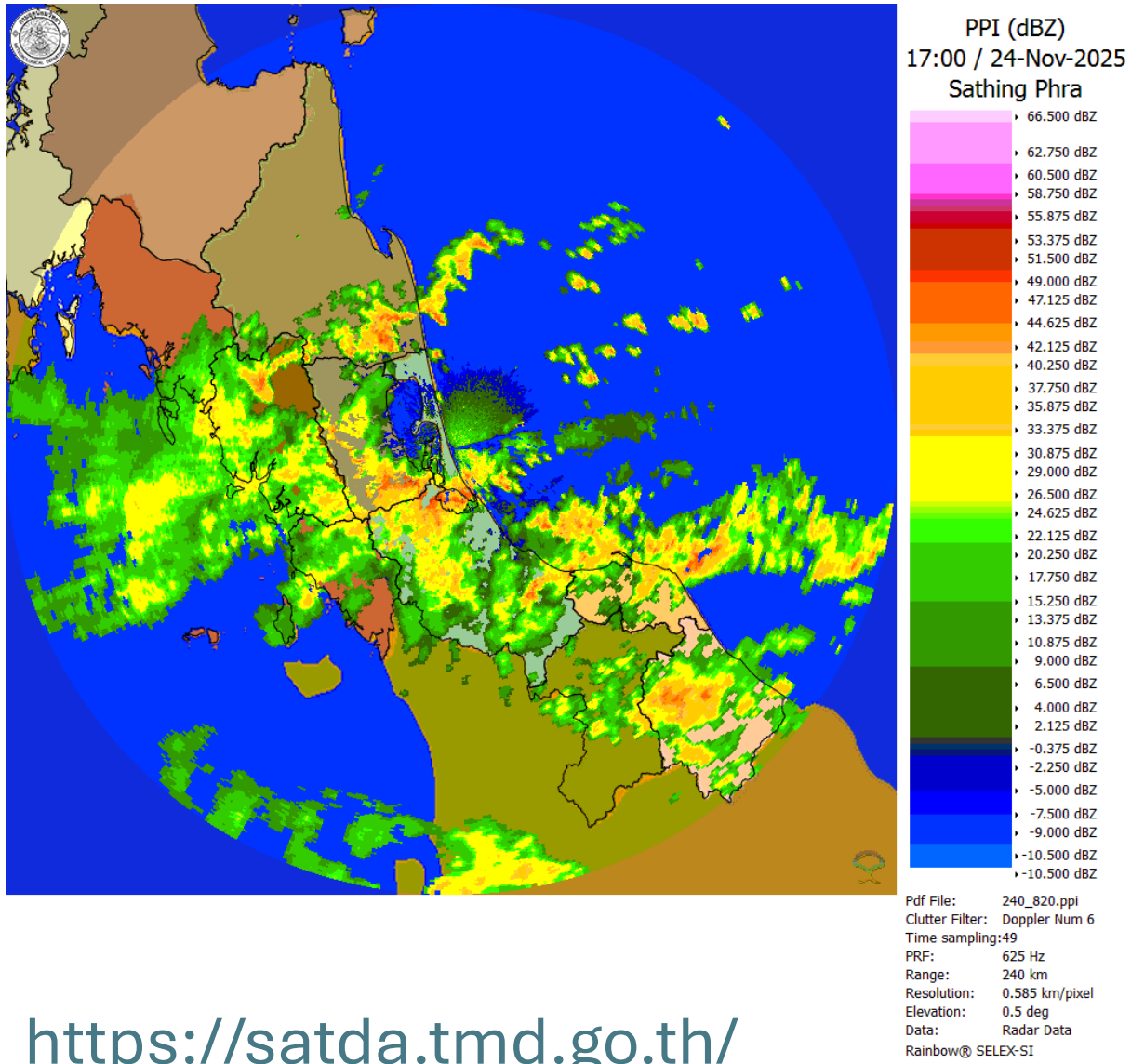

Waipa Gardens Accumulation
Kdp Derived
Radar Derived
Lightning Derived
22.25° N
22.20°
22.15°
159.55° W
159.5°
159.45°
Rainfall Rate (cm/hr)
Rainfall Accumulation (cm)
Date-Time (UTC)
0
5
10
15
20
25
30
0
20
40
60
80
100
120
140
14-2145
2345
15-0145
0345
0545
0745
0945
1145
1345
1545
1745
1945
2145
2345

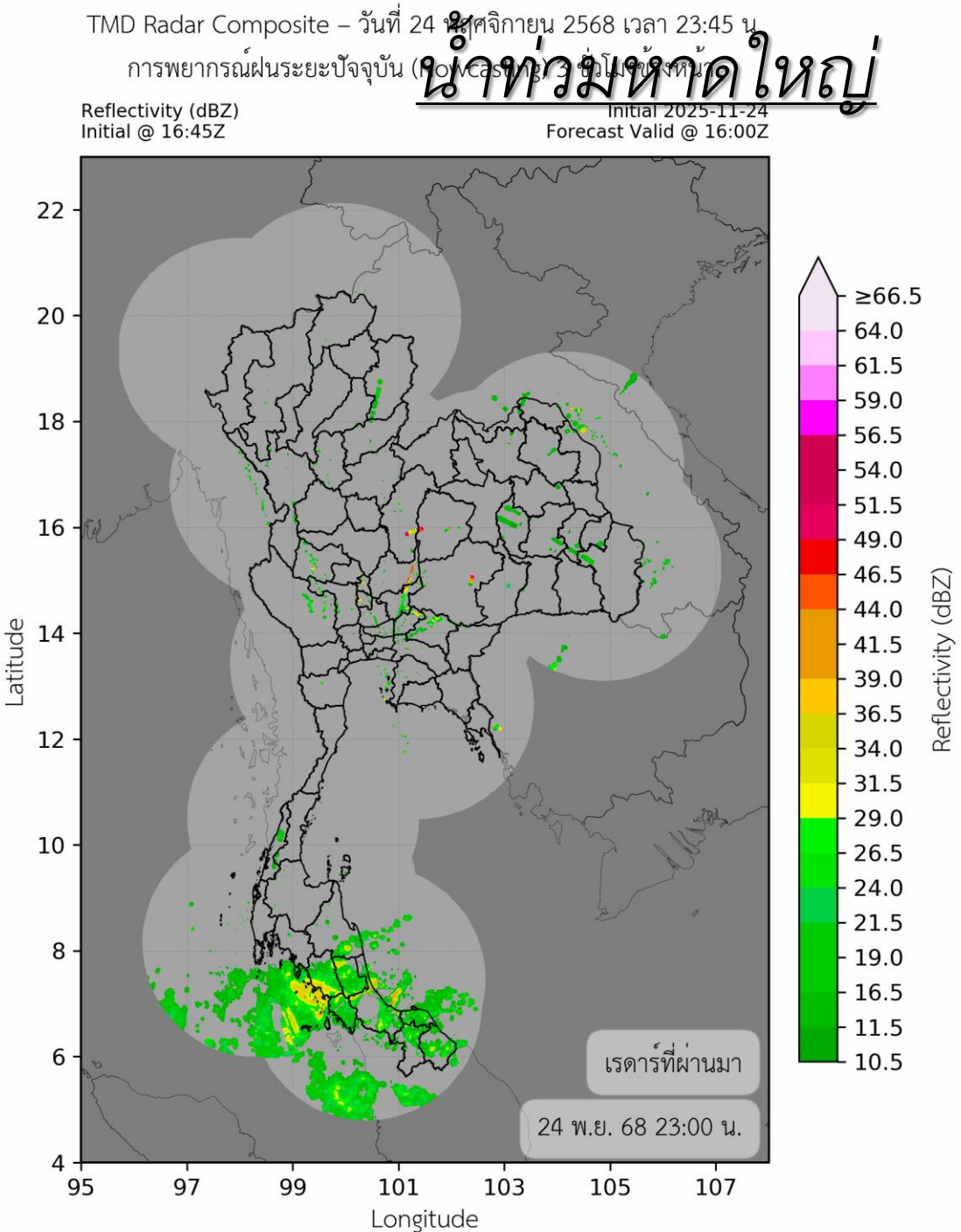


กรมอุตุนิยมวิทยา กระทรวงดิจิทัลเพื่อเศรษฐกิจและสังคม
Thai Meteorological Department, Ministry of Digital Economy and Society



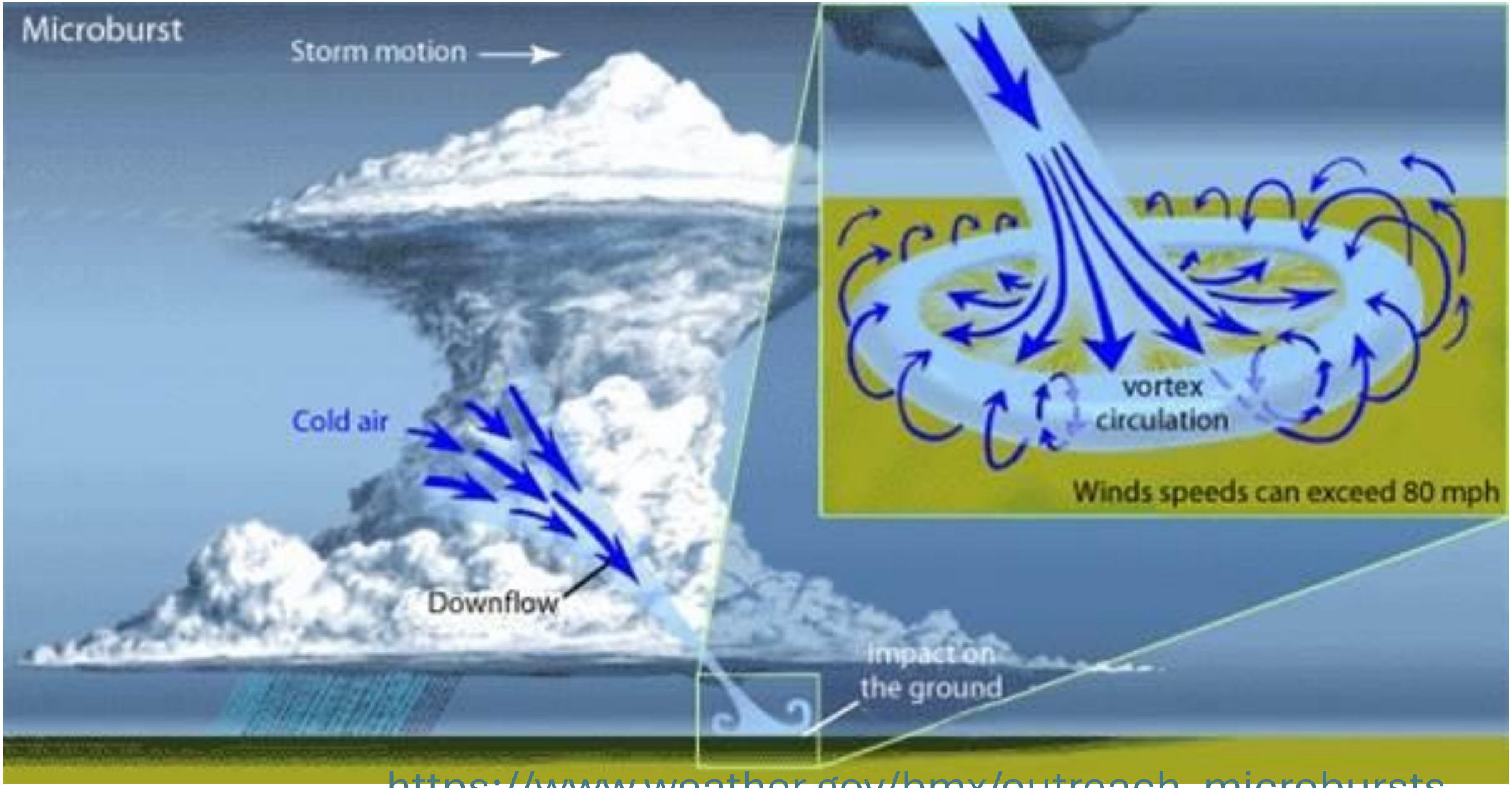
น้ำท่วมหาดใหญ่





https://satda.tmd.go.th/wp-

Microburst.



https://www.weather.gov/bmx/outreach_microbursts

https://www.weather.gov/bmx/outreach_microbursts

microburst

A convective downdraft (downburst) that covers an area less than 4 km along a side with peak winds that last 2–5 minutes.

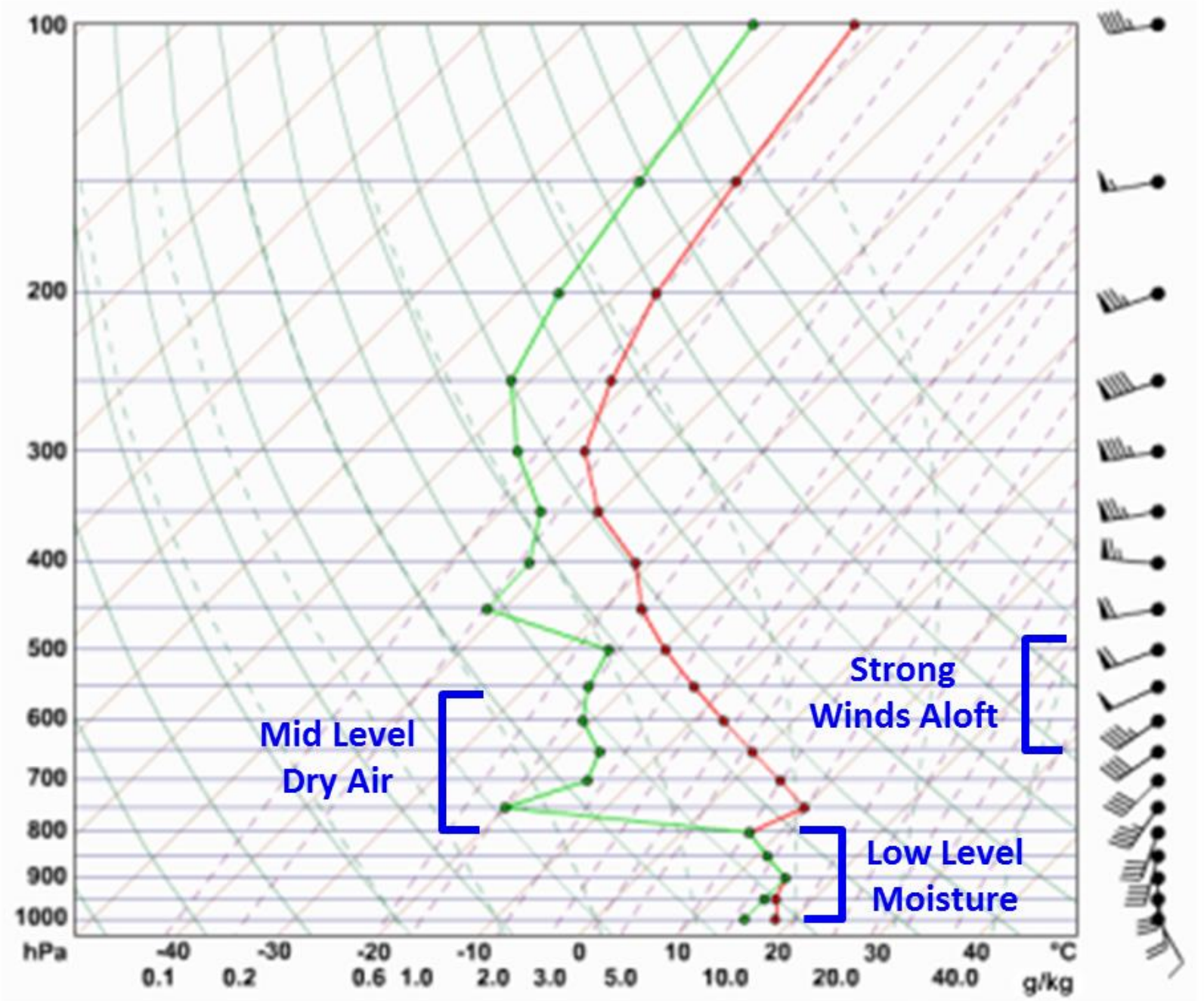
Differential velocity across the divergence center is greater than 10 m s^{-1}. The strong wind shears associated with a microburst can result in aircraft accidents. Microbursts can be wet (contain rain) or dry.

Atkins, N. T., and R. M. Wakimoto, 1991: Wet microburst activity over the southeastern United States: Implications for forecasting. *Wea. Forecasting*, **6**, 470–482, https://doi.org/10.1175/1520-0434(1991)006<0470:WMAOTS>2.0.CO;2.

Term edited 22 June 2018.

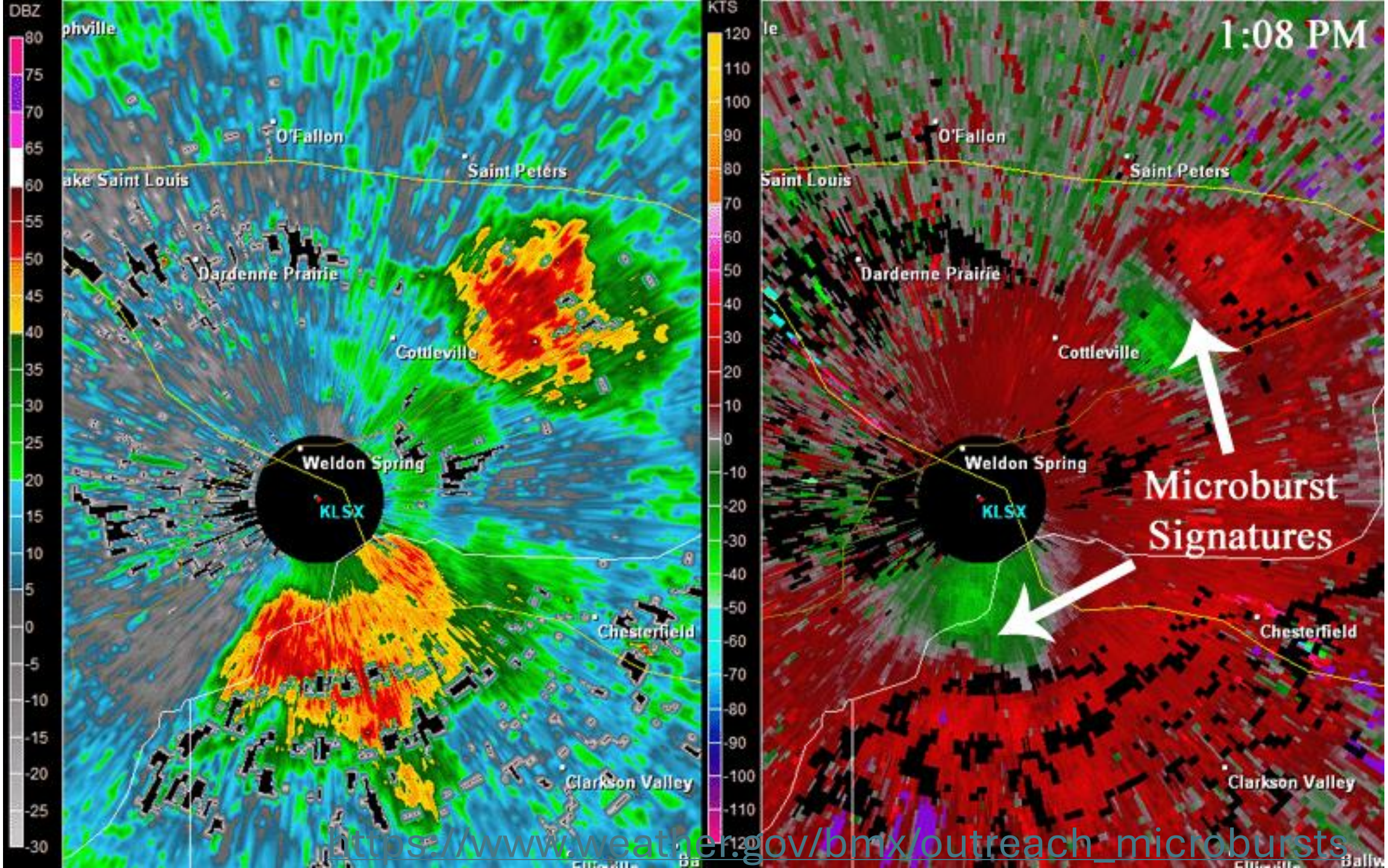
Copyright 2026 American Meteorological Society (AMS). For permission to reuse any portion of this work, please contact permissions@ametsoc.org. Any use of material in this work that is determined to be "fair use" under Section 107 of the U.S. Copyright Act (17 U.S. Code § 107) or that satisfies the conditions specified in Section 108 of the U.S.Copyright Act (17 USC § 108) does not require AMS's permission. Republication, systematic reproduction, posting in electronic form, such as on a website or in a searchable database, or other uses of this material, except as exempted by the above statement, require written permission or a license from AMS. Additional details are provided in the AMS Copyright Policy statement.

https://glossary.ametsoc.org/wiki/microburst/



https://www.weather.gov/bmx/outreach_microbursts

©The COMET Program



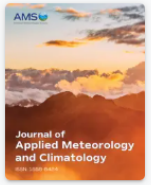
https://www.weather.gov/bmx/outreach_microbursts

ตัวอย่างเปเปอร์งานวจัย

Microburst.

AMS American Meteorological Society | Journals

JOURNALS BROWSE PUBLISH SUBSCRIBE ABOUT



Journal of Applied Meteorology and Climatology

Volume 23: Issue 6

Cited By

Publications Citing This Document
You are looking at 1-50 of 270 citations

Next >

[1] Liu, Xinying, Abà, Anna, Capone, Pierluigi, Manfriani, Leonardo, and Fu, Yongling, 2022, "Atmospheric Disturbance Modelling for a Piloted Flight Simulation Study of Airplane Safety

< Previous Article Next Article >

Editorial Type: Article

Article Type: Research Article

Microburst Wind Structure and Evaluation of Doppler Radar for Airport Wind Shear Detection

James W. Wilson, Rita D. Roberts, Cathy Kessinger, and John McCarthy

Print Publication: 01 Jun 1984

DOI: https://doi.org/10.1175/1520-0450(1984)023<0898:MWSAEO>2.0.CO;2

Page(s): 898–915

Cited by: 270

Article History | Download PDF | © Get Permissions

Abstract/Excerpt | Full Text | PDF

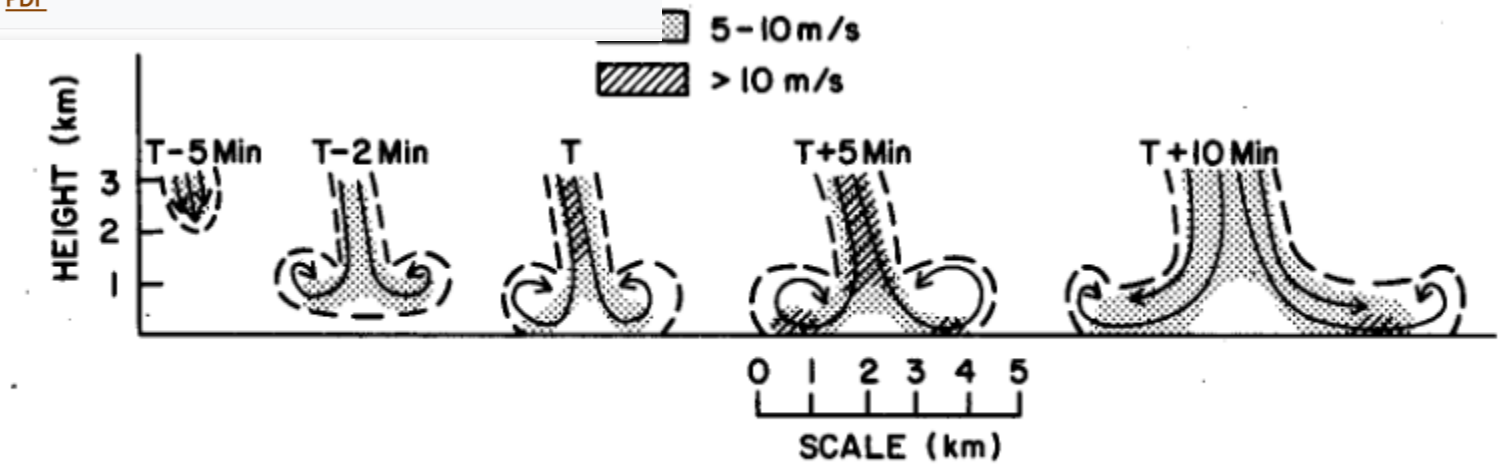


FIG. 6. Vertical cross section of the evolution of the microburst wind field based on a summary of data in Section 3; T is the time of initial divergence at the surface. The shading denotes the vector wind speeds.

https://journals.ametsoc.org/view/journals/apme/23/6/1520-0450_1984_023_0898_mwsaeo_2_0_co_2.xml